

BIO-INSPIRED AUTONOMOUS BUILDING HEALTH AND ENERGY MANAGEMENT SYSTEM USING MACHINE LEARNING AND ARTIFICIAL IMMUNE SYSTEMS



Project Report Submitted For

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Abstract

Modern smart infrastructure systems require intelligent monitoring frameworks capable of identifying environmental, structural, and operational anomalies in real time. Traditional monitoring systems rely primarily on static thresholds and isolated parameter analysis, resulting in poor adaptability, high false alarm rates, and inefficient decision-making under dynamic building conditions. To overcome these limitations, this project proposes a bio-inspired autonomous building health and energy management framework integrating artificial immune system principles with machine learning techniques for adaptive anomaly detection and intelligent monitoring [3].

The proposed framework combines synthetic multi-sensor data generation, preprocessing, Isolation Forest anomaly detection[1], Autoencoder-based deep learning [2], multi-factor fusion, adaptive learning, and autonomous decision-making within a unified software-based architecture. Environmental and structural parameters including temperature, humidity, energy consumption, occupancy, structural strain, and vibration are continuously analyzed to identify complex fault conditions. The hybrid anomaly detection framework combines rule-based reasoning with machine learning models to improve robustness and reduce false alarms.

Experimental evaluation demonstrated strong anomaly detection capability with high classification performance and adaptive learning efficiency. The framework achieved an ROC-AUC score of 0.89, indicating excellent anomaly discrimination capability. Results also demonstrated improved multi-condition analysis and reduced false alarm generation. The project highlights the effectiveness of combining artificial immune systems with machine learning for intelligent smart infrastructure monitoring and adaptive autonomous management systems [1], [2],[3].

1 Introduction

The rapid advancement of smart infrastructure technologies has increased the need for intelligent monitoring systems capable of ensuring operational reliability, structural safety, and energy efficiency. Modern buildings consist of interconnected environmental control systems, occupancy management frameworks, HVAC systems, structural monitoring sensors, and intelligent energy management platforms. Continuous monitoring of these systems is critical to prevent structural degradation, operational inefficiency, and excessive energy consumption [5], [6].

Traditional smart building monitoring systems primarily depend on static threshold-based analysis. In such systems, alerts are generated whenever sensor values exceed predefined limits. Although simple to implement, these approaches suffer from several major limitations. Static thresholds cannot adapt effectively under dynamic environmental conditions and often produce unnecessary alarms or fail to identify hidden anomalies. Furthermore, most conventional systems analyze parameters independently, making them ineffective for identifying correlated abnormal conditions.

In real-world infrastructure environments, abnormal conditions rarely occur as isolated events. For example, high vibration combined with abnormal structural strain may indicate structural instability, while increased temperature combined with excessive energy consumption may indicate HVAC malfunction. Single-parameter monitoring systems cannot effectively identify such correlated faults.

Recent advancements in artificial intelligence and machine learning have enabled the development of intelligent anomaly detection systems capable of adaptive learning and nonlinear pattern recognition [4]. Isolation Forest algorithms isolate abnormal samples using random partitioning trees [1], while Autoencoder neural networks learn compressed representations of normal behavior and identify anomalies through reconstruction error analysis [2].

Artificial immune systems introduce another dimension of intelligent monitoring by mimicking biological immune behavior [3]. Biological immune systems continuously detect harmful elements, generate responses, and store memory for future adaptation. Applying these principles to smart infrastructure monitoring enables continuous learning, adaptive anomaly analysis, and autonomous decision-making.

2 Literature Review

Several intelligent anomaly detection systems have been proposed for smart infrastructure monitoring in recent years. Traditional building monitoring systems mainly relied on static thresholds and rule-based alert generation. However, these approaches lack adaptability and produce excessive false alarms under dynamic operating conditions.

Liu et al. proposed the Isolation Forest algorithm for unsupervised anomaly detection using random partitioning trees [1]. Isolation Forest became popular because of its low computational complexity and ability to isolate abnormal samples without requiring labeled datasets. However, standalone Isolation Forest models may struggle to capture highly nonlinear environmental relationships.

Sakurada et al. introduced Autoencoder-based anomaly detection systems capable of learning compressed representations of normal behavior [2]. Autoencoders identify anomalies using reconstruction error analysis and perform effectively for nonlinear pattern recognition. Although Autoencoders demonstrate strong anomaly detection capability, standalone deep learning models may become unstable under highly dynamic conditions if not integrated with additional reasoning systems.

Dasgupta proposed artificial immune systems inspired by biological immune mechanisms [3]. Artificial immune systems perform adaptive learning, detection, memory formation, and autonomous response generation. These systems have been applied in cybersecurity and industrial fault diagnosis but remain relatively underexplored in intelligent infrastructure monitoring applications.

Chen et al. developed machine learning-based structural health monitoring systems using vibration analysis and anomaly detection [5]. Their work demonstrated the effectiveness of intelligent monitoring for civil infrastructure systems. However, adaptive memory learning and autonomous reasoning mechanisms were not fully integrated.

Khan et al. proposed deep learning-based smart building monitoring systems using neural networks and environmental data analysis [6]. The framework achieved strong classification accuracy but lacked multi-condition fusion and adaptive immune-inspired learning capability.

Existing intelligent monitoring systems still suffer from several limitations such as Independent parameter analysis ,Poor adaptive learning capability ,High false alarm generation ,Lack of autonomous intelligence and Limited multi-condition anomaly detection.

The proposed framework addresses these limitations by integrating: Artificial immune systems, Isolation Forest, Autoencoder deep learning, Multi-factor fusion and Adaptive memory learning within a unified intelligent architecture.

3 Problem Statement

Modern smart infrastructure systems generate large amounts of environmental, structural, and operational data continuously. Traditional monitoring systems rely on static threshold-based analysis and isolated parameter monitoring, making them ineffective for identifying correlated abnormal conditions and adaptive environmental variations.

These limitations result in:

- High false alarm rates
- Delayed anomaly detection
- Poor adaptability
- Lack of autonomous intelligence
- Inefficient energy management
- Increased maintenance complexity

Real-world abnormal conditions often involve multiple correlated parameters rather than isolated sensor deviations. Existing systems fail to identify such complex multi-condition anomalies effectively.

Therefore, there is a need for an intelligent adaptive monitoring framework capable of Continuous learning ,Multi-factor anomaly analysis ,Autonomous decision-making ,Adaptive memory-driven optimization and Reduced false alarm generation.

The proposed project addresses these challenges using bio-inspired artificial immune systems integrated with machine learning-based anomaly detection techniques.

The major objectives of the project are:

- To design a bio-inspired intelligent monitoring framework
- To simulate realistic building sensor data

- To implement adaptive machine learning-based anomaly detection
- To reduce false alarm generation
- To enable multi-condition anomaly analysis
- To perform autonomous intelligent decision-making
- To incorporate adaptive memory-driven learning

4 Proposed System Architecture

The proposed framework follows a multi-layer intelligent architecture inspired by biological immune systems.

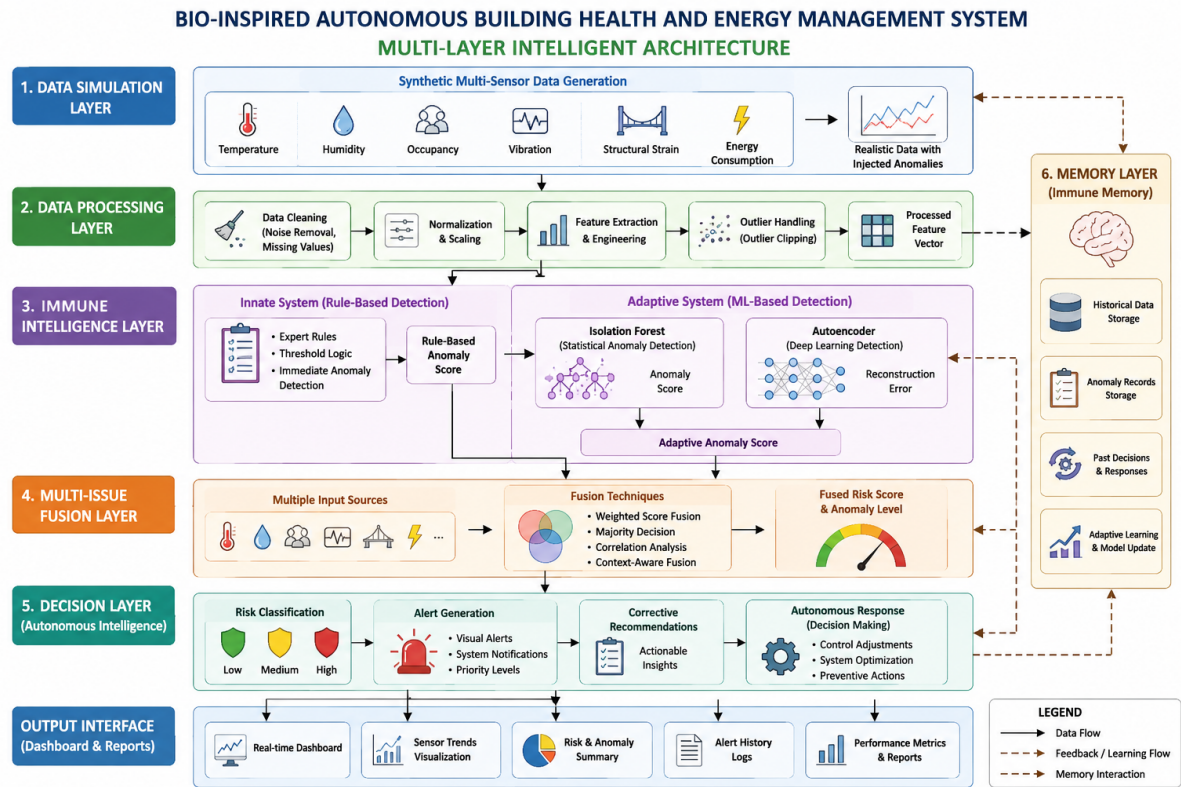


Figure 1: Architecture diagram of the system.

The architecture consists of:

- Data Simulation Layer
- Data Processing Layer

- Immune Intelligence Layer
- Multi-Issue Fusion Layer
- Decision Layer
- Memory Layer

The Data Simulation Layer generates realistic synthetic sensor streams representing temperature, humidity, occupancy, structural strain, structural vibration, and energy consumption.

The Data Processing Layer performs normalization, scaling, noise reduction, and feature extraction.

The Immune Intelligence Layer contains:

- Innate System (Rule-Based)
- Adaptive System (Machine Learning)

The Innate System performs immediate anomaly analysis using predefined expert conditions, while the Adaptive System integrates Isolation Forest and Autoencoder models for adaptive anomaly detection.

The Multi-Issue Fusion Layer combines outputs from multiple models and sensor streams to identify correlated fault conditions.

The Decision Layer generates alerts, risk classifications, and corrective recommendations.

The Memory Layer stores historical anomalies and system behavior for continuous adaptive learning.

5 Methodology

The methodology adopted in this project focuses on developing a fully software-based intelligent monitoring framework capable of adaptive anomaly detection and autonomous decision-making.

The proposed project introduces a bio-inspired autonomous building health and energy management framework integrating:

- Artificial immune systems

- Isolation Forest anomaly detection
- Autoencoder deep learning
- Multi-factor fusion
- Adaptive memory learning
- Autonomous decision-making

within a unified software-based architecture.

The project demonstrates that combining artificial immune systems with machine learning significantly improves anomaly detection capability and intelligent infrastructure monitoring performance.

The first stage involves synthetic sensor data generation. Since the framework is software-based, realistic environmental and structural datasets were simulated using Python-based statistical generation techniques.

Controlled anomalies were injected into the data to simulate realistic abnormal conditions including HVAC malfunction, structural instability, environmental imbalance, and excessive energy consumption.

The preprocessing stage included:

- Missing value handling
- Feature normalization
- Noise reduction
- Outlier clipping
- Scaling

Feature engineering techniques derived secondary indicators such as occupancy-adjusted energy usage and vibration-strain correlation.

Isolation Forest anomaly detection [1] was implemented for unsupervised anomaly analysis using random partitioning trees.

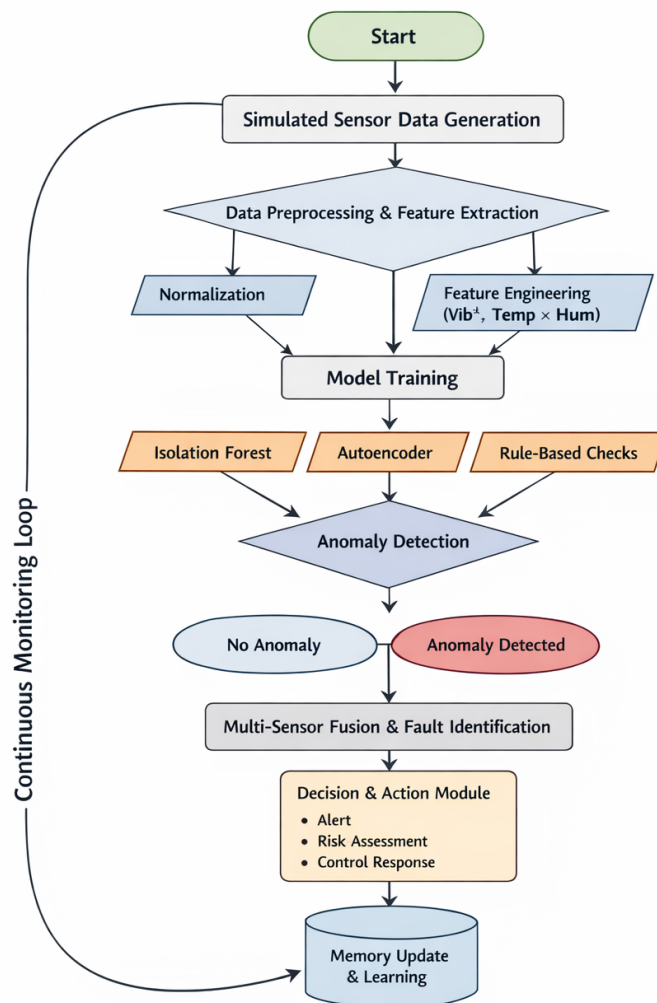
$$Score(x, n) = 2^{-\frac{E(h(x))}{c(n)}}$$

Autoencoder neural networks [2], [4] were implemented for nonlinear anomaly detection using reconstruction error analysis.

$$Loss = ||x - \hat{x}||^2$$

Outputs from Isolation Forest, Autoencoder, and rule-based reasoning were combined using weighted anomaly fusion techniques within the Multi-Issue Fusion Layer.

The Memory Layer continuously stored historical anomalies and previous decisions to improve future classification capability through adaptive learning [3].



Bio-Inspired SHM System Methodology

Figure 2: Flow Chart

6 Implementation Details

The proposed framework was implemented entirely using Python-based simulation and machine learning environments.

The software tools used include:

- Python
- NumPy [11]
- Pandas [10]
- Scikit-learn [9]
- TensorFlow [8]
- Matplotlib
- Streamlit [13]

Isolation Forest was implemented using Scikit-learn with:

- 100 estimators
- 0.05 contamination factor
- Randomized partition trees

The Autoencoder architecture consisted of encoder layers, bottleneck compression layers, and decoder layers.

Training configuration included:

- Epochs: 100
- Optimizer: Adam
- Batch size: 32
- Mean Squared Error loss

Dropout regularization and early stopping techniques improved generalization capability and prevented overfitting [4].

The Streamlit dashboard provided real-time visualization, risk indicators, performance graphs, and alert notifications.

7 Results and Discussion

The proposed framework was evaluated using synthetic multi-sensor datasets containing both normal and abnormal operating conditions.

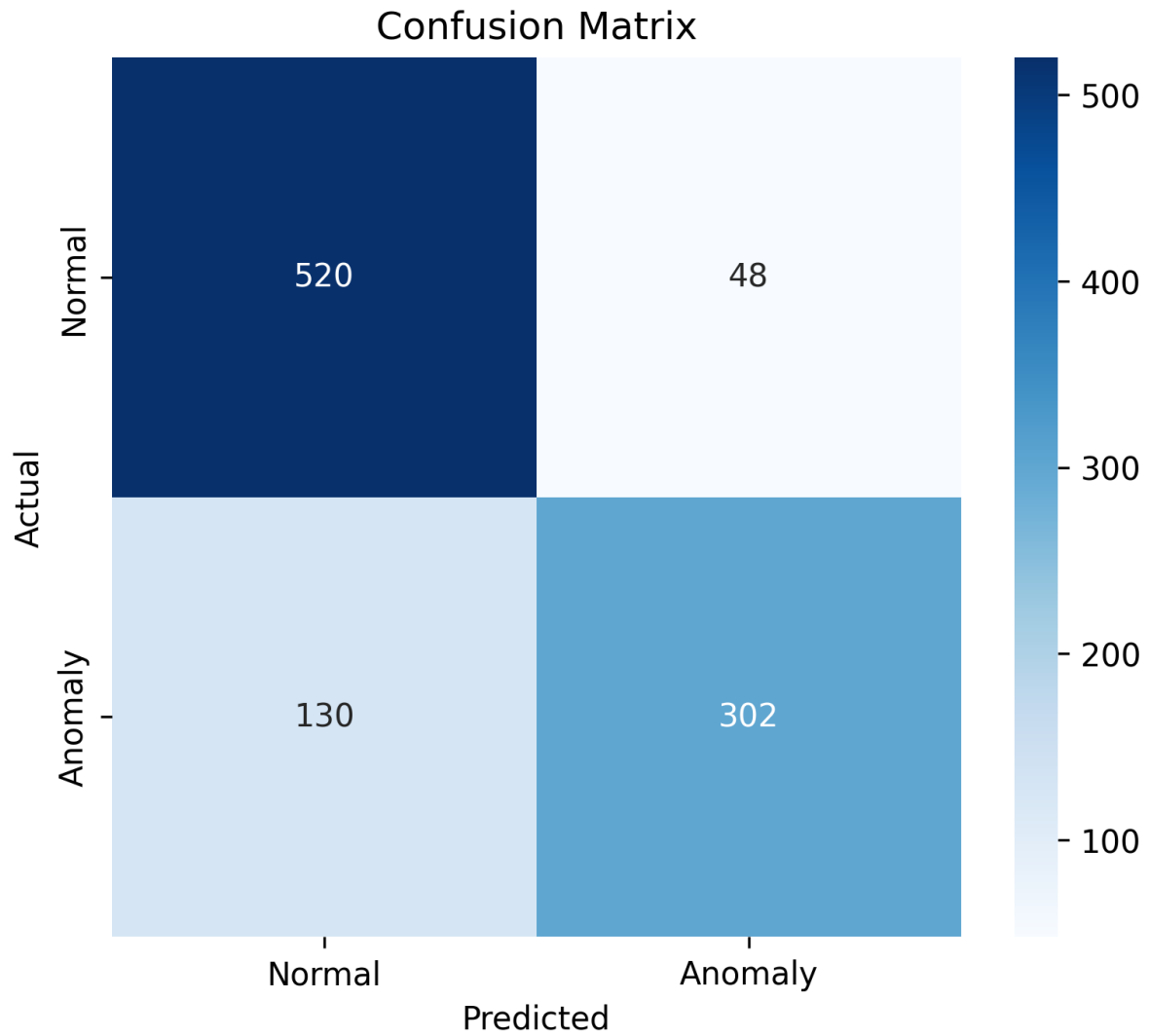


Figure 3: Confusion Matrix

The confusion matrix values were:

Actual / Predicted	Normal	Anomaly
Normal	520	48
Anomaly	130	302

Table 1: Confusion Matrix Values

The system correctly classified 520 normal samples and 302 anomalous samples.

The low false positive count of 48 demonstrates strong stability and reduced unnecessary alert generation. The false negative count of 130 indicates that some anomalies remained undetected. However, the majority of anomalies were correctly identified, demonstrating strong anomaly discrimination capability.

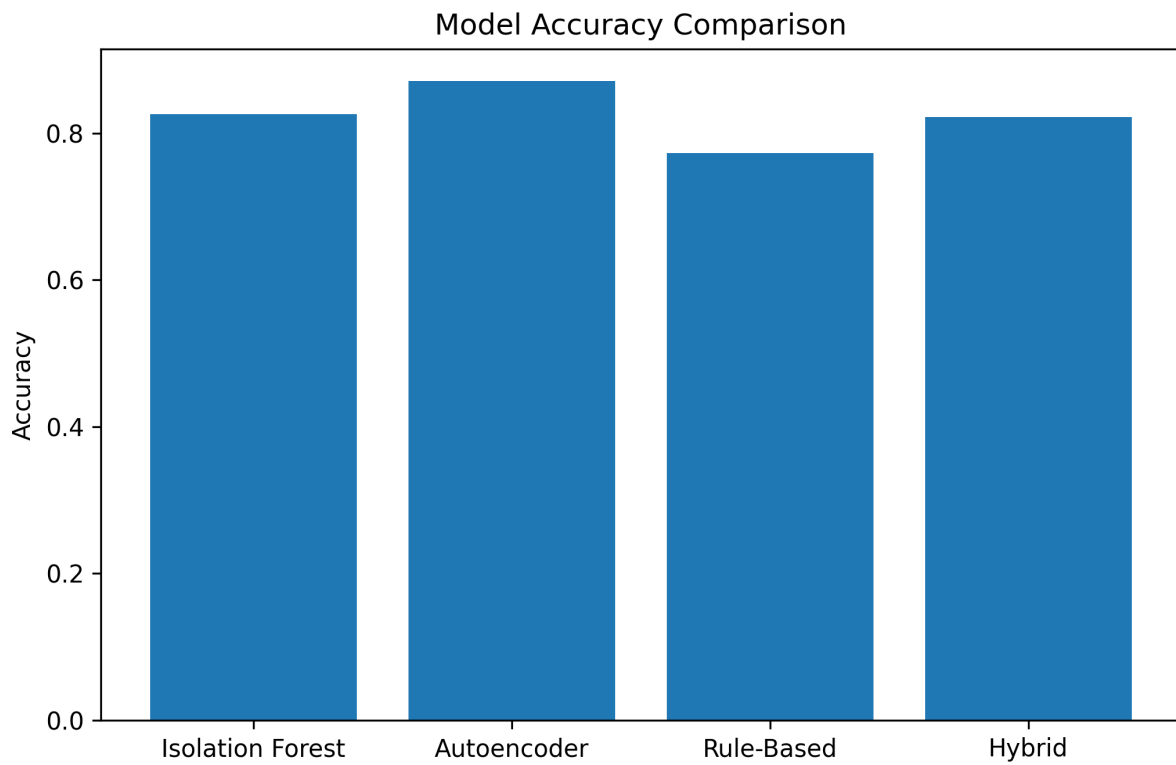


Figure 4: Model Accuracy Comparison

The accuracy comparison graph evaluated different anomaly detection models.

Model	Accuracy
Isolation Forest	82.5%
Autoencoder	87.2%
Rule-Based	77.4%
Hybrid	82.3%

Table 2: Accuracy Comparison

The Autoencoder achieved the highest standalone accuracy because deep neural networks effectively capture nonlinear feature relationships [2], [4].

Isolation Forest demonstrated strong anomaly isolation capability because abnormal samples require fewer partitions for separation [1].

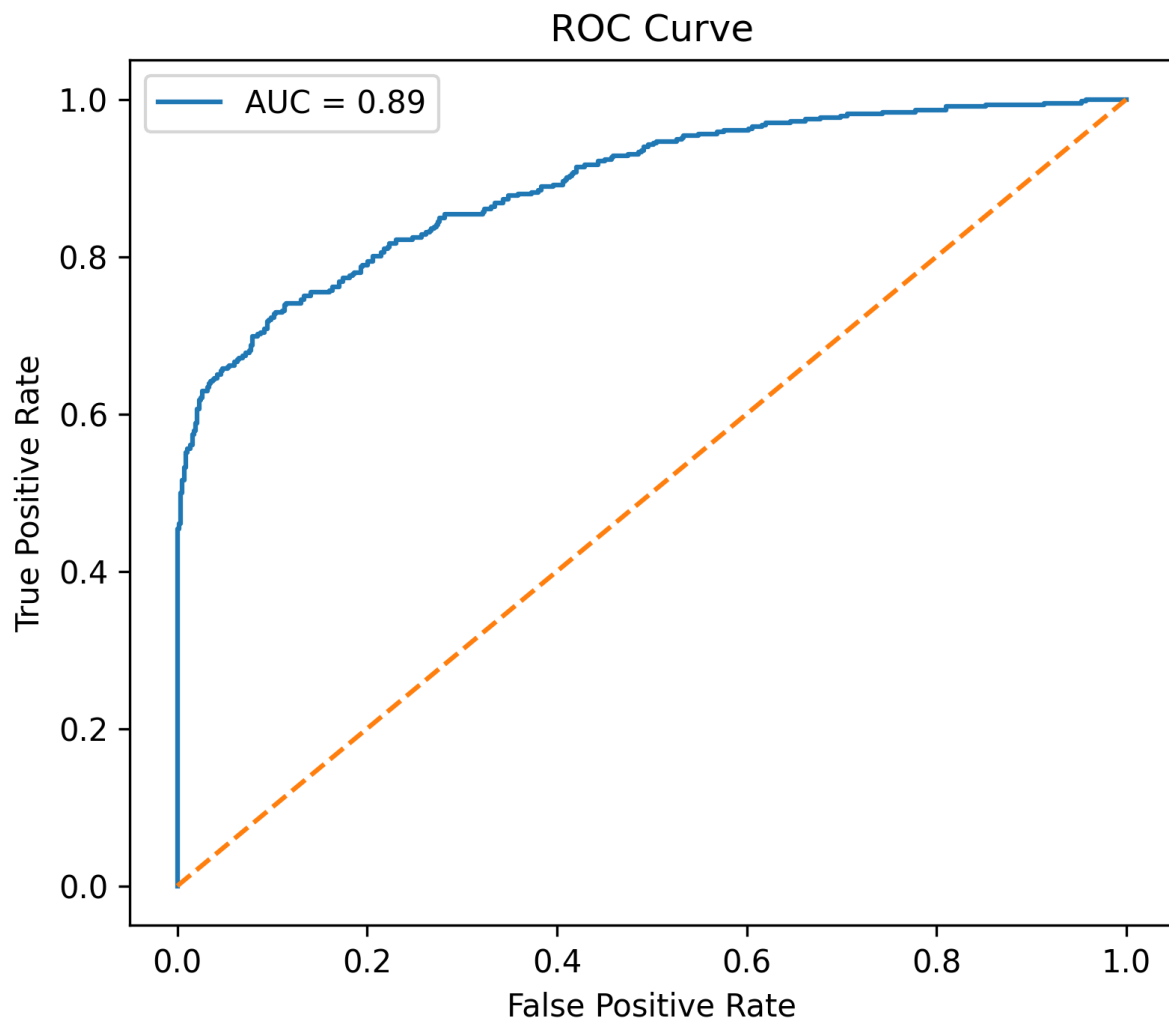


Figure 5: ROC Curve

The ROC curve evaluated classification capability under varying threshold conditions.

$$AUC = 0.89$$

An AUC score close to 1 indicates excellent classification capability.

The ROC curve demonstrates:

- High sensitivity
- Low false positive rates
- Strong anomaly discrimination

- Stable classification performance

The experimental results confirm that combining machine learning with artificial immune systems significantly improves intelligent monitoring capability.

8 Performance Analysis

The proposed framework demonstrated strong anomaly detection performance with reduced false alarm generation and improved adaptive learning capability.

Key performance metrics included Accuracy ,Precision ,Recall ,F1-score and ROC-AUC.

Table 3: Model Performance Evaluation Metrics

Metric	Value
Accuracy	0.822000
Precision	0.862857
Recall	0.699074
F1-Score	0.772379
ROC-AUC	0.891126
False Alarm	0.084507

The hybrid framework improved overall classification stability because anomaly decisions were validated across Isolation Forest, Autoencoder, and rule-based reasoning before final classification.

The framework successfully handled environmental anomalies, structural anomalies, energy abnormalities, and multi-condition correlated faults.

Compared to traditional threshold systems, the proposed framework demonstrated:

- Better adaptability
- Higher intelligence
- Reduced false alarms
- Improved robustness
- Continuous learning capability

9 Conclusion

This project successfully developed a bio-inspired autonomous building health and energy management framework integrating artificial immune systems, Isolation Forest anomaly detection, Autoencoder deep learning, multi-factor fusion, adaptive memory learning, and autonomous decision-making within a unified intelligent architecture.

Traditional threshold-based systems suffer from poor adaptability and high false alarm generation. The proposed framework addressed these limitations through adaptive machine learning and immune-inspired intelligence [3].

Experimental evaluation demonstrated strong anomaly detection capability, reduced false alarms, effective nonlinear learning, multi-condition anomaly analysis, and autonomous intelligent behavior.

The ROC-AUC score of 0.89 confirmed strong classification performance and effective anomaly discrimination capability.

The project demonstrates that combining artificial immune systems with machine learning provides an effective solution for next-generation intelligent infrastructure monitoring systems.

10 Future Work

Future enhancements may include:

- Real-time IoT sensor integration
- Edge AI deployment
- Reinforcement learning
- Transformer-based anomaly detection
- Explainable AI integration
- Cloud analytics
- Digital twin implementation
- Federated learning

- Smart city infrastructure monitoring
- Predictive maintenance systems
- Autonomous optimization frameworks

The proposed architecture provides a scalable foundation for future intelligent infrastructure management systems.

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